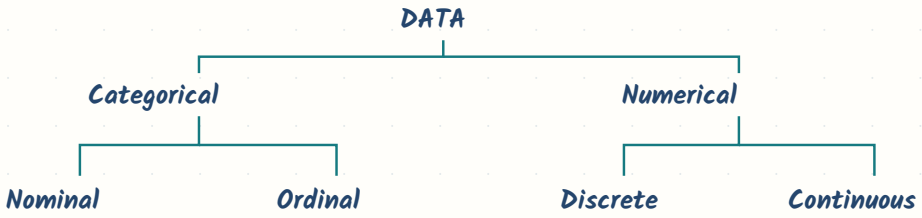
Statistics · data & central tendency

Statistics is the study of the collection, analysis, interpretation, presentation and organization of data.

Types of data



Categorical (qualitative): categories such as colour or yes/no. These describe qualities rather than measured amounts.

Nominal: no ordering. One colour is not inherently better than another; categories cannot be ranked.

Ordinal: ordered categories, e.g. health status: good > OK > poor.

Numerical (quantitative): values that can be counted or measured, e.g. height and weight.

Discrete: distinct, countable values; e.g. number of students.

Continuous: measured values within an interval; e.g. height.

Interval scale: equal intervals, but no meaningful absolute zero; values may be negative. Example: temperature in °C.

Ratio scale: equal intervals and a meaningful zero. Examples: age, distance and length.

Measures of central tendency

A single number describes the centre or typical value of a set of scores. Three common measures: mean, median and mode.

Mean: the arithmetic average.

$$\bar{x} = \Sigma x_i \,/\, N$$

Example: 1, 2, 3, 4, 5 → $\bar{x} = (1+2+3+4+5)/5 = 3$.

For frequency data, multiply each value by its frequency:

$$\bar{x} = \Sigma(f_i x_i) \,/\, \Sigma f_i$$

x	1	2	3
f	2	1	3

Raw data: 1, 1, 2, 3, 3, 3.

Mean = $(1\times2 + 2\times1 + 3\times3)/6 = 13/6 \approx 2.167$.

Notation: $\bar{x}$ for the sample mean; μ for the population mean. A sample is a subset of a population.

Mean, median, mode & variability

Mean · useful properties

The mean is widely used and includes every data value. Extreme scores can affect it strongly.

The sum of the deviations from the mean is zero:

$$\Sigma(x_i - \bar{x}) = 0$$

Use the mean when calculating standard deviation, coefficient of variation and skewness, or when a stable measure using all values is desired.

Median

The median divides ordered data into two equal parts.

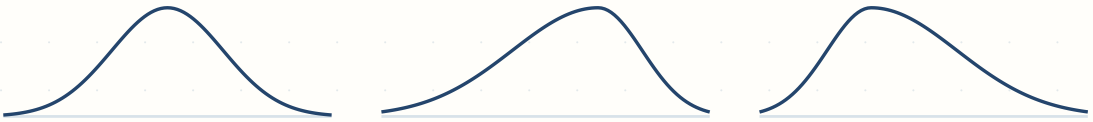
Sort the data and find the centre value. For an even number of observations, average the two central values.

Mode

The mode is the value with the highest frequency. It applies to both qualitative and quantitative data.

Unimodal: one mode. Bimodal: two values share the highest frequency. Multimodal: more than two modes.

Symmetric & asymmetric data



Symmetric Left skew Right skew

For a symmetric unimodal distribution: mean = median = mode.

Negative (left) skew: mean < median. Positive (right) skew: mean > median.

Pearson's coefficients of skewness:

$$(\text{mean} - \text{mode}) / SD$$

$$\text{or } 3(\text{mean} - \text{median}) / SD$$

Measures of variability

Range, variance and standard deviation describe the spread of scores and their distance from the mean.

Variability helps show how well an individual score represents the distribution.

Range, standard deviation & quartiles

Range

The distance from the smallest value to the largest:

range = maximum − minimum

Standard deviation

A common measure of variability: the typical distance of values from the mean, computed from squared deviations.

Deviation = $x - \mu$. Variance is the average squared deviation; standard deviation is its positive square root.

Population:

$$\sigma^2 = \Sigma(x_i - \mu)^2 / N$$

$$\sigma = \sqrt{[\Sigma(x_i - \mu)^2 / N]}$$

Sample:

$$s^2 = \Sigma(x_i - \bar{x})^2 / (n - 1)$$

$$s = \sqrt{[\Sigma(x_i - \bar{x})^2 / (n - 1)]}$$

For sample notation: σ becomes s, μ becomes $\bar{x}$, and the denominator is $n-1$.

Worked example

Data: 1, 5, 7, 8, 9. The mean is $\mu = 6$.

x	1	5	7	8	9
$x - \mu$	−5	−1	1	2	3
$(x - \mu)^2$	25	1	1	4	9

Sum of squared deviations, $SS = 25+1+1+4+9 = 40$.

$$\sigma^2 = 40/5 = 8;\ \sigma = \sqrt{8} \approx 2.828$$

For these values treated as a sample: $s^2 = 40/4 = 10$.

The mean and standard deviation together help describe the data.

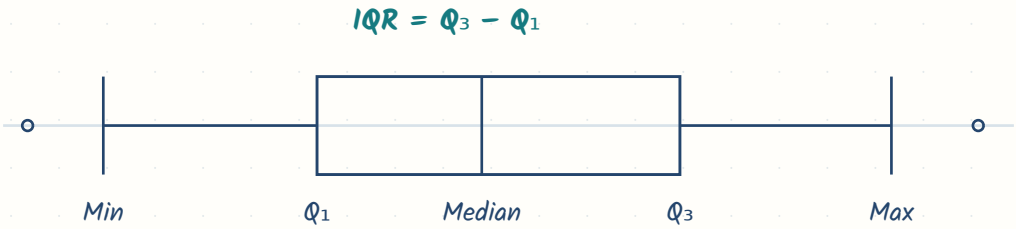
Five-number summary

Minimum, Q_1, median (Q_2), Q_3 and maximum.

interquartile range (IQR) = $Q_3 - Q_1$

Box plots & probability

Box-and-whisker plot



The box spans Q_1 to Q_3; the line within it marks the median.

Whiskers extend to the most extreme observations within the fences.

lower fence = Q_1 − 1.5 IQR

upper fence = Q_3 + 1.5 IQR

Values outside these fences are potential outliers. Values beyond $Q_1 - 3$ IQR or $Q_3 + 3$ IQR are sometimes called extreme outliers.

Probability · basic language

A random experiment is an action whose outcome is not known in advance. Example: tossing a coin.

The sample space S contains all possible outcomes. An event is a subset of S.

Classical probability (equally likely outcomes):

P(A) = favourable outcomes / total outcomes

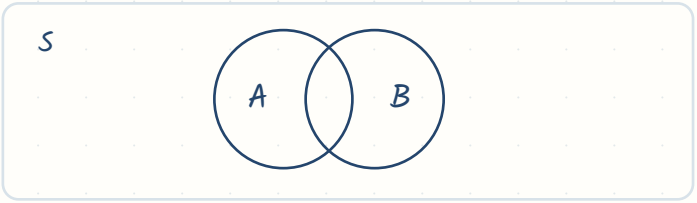
Example: P(head) = 1/2.

Empirical probability: observed occurrences / number of trials.

Example: 450 heads in 1,000 tosses → 450/1000 = 0.45.

$P(S) = 1$; $0 \le P(A) \le 1$

Union, intersection & exclusive events



$P(A \cup B) = P(A) + P(B) - P(A \cap B)$

Mutually exclusive events have no outcomes in common: $A \cap B = \varnothing$, so $P(A \cap B) = 0$. They cannot occur together.

For mutually exclusive events: $P(A \cup B) = P(A) + P(B)$.

Probability rules & conditioning

Independent events

One event does not affect the probability of the other. Independent draws with replacement are a typical example.

$$P(A \cap B) = P(A)P(B)$$

Complements

A^c is the event that A does not happen.

$$n(A^c) = n(S) - n(A)$$

$$P(A^c) = 1 - P(A)$$

$$P(A) + P(A^c) = 1$$

Here $n(\cdot)$ counts outcomes, whereas $P(\cdot)$ denotes probability.

$$P(A \cap B^c) = P(A) - P(A \cap B)$$

$$P(A^c \cap B) = P(B) - P(A \cap B)$$

Addition rule

The probability that A or B occurs is:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

For mutually exclusive A and B, omit the intersection term.

Conditional probability

$P(A \mid B)$ means the probability of A given that B has happened. B is the conditioning event.

$$P(A \mid B) = P(A \cap B) / P(B),\ P(B) > 0$$

Multiplication rule

$$P(A \cap B) = P(A \mid B)P(B)$$

$$\text{also} = P(B \mid A)P(A),\ P(A) > 0$$

For multiple events, multiply successive conditional probabilities:

$$P(A_1 \cap \ldots \cap A_n)$$

$$= P(A_1)\ P(A_2 \mid A_1)\ \ldots\ P(A_n \mid A_1 \cap \ldots \cap A_{n-1})$$

Independence through conditioning

If B has no effect on A, then $P(A \mid B) = P(A)$.

Substituting into the multiplication rule gives $P(A \cap B) = P(A)P(B)$. This equality also characterizes independence.

Total probability, Bayes & Naive Bayes

Three-event addition rule

$$P(A \cup B \cup C) = P(A)+P(B)+P(C)$$
$$- P(A \cap B) - P(A \cap C) - P(B \cap C)$$
$$+ P(A \cap B \cap C)$$

Law of total probability

Events A_1, A_2, ..., A_k are exhaustive if their union is the entire sample space S. They are pairwise mutually exclusive if no two overlap.

If they partition S, then B can be split into disjoint parts $B \cap A_i$.

$$P(B) = \sum_i P(B \cap A_i)$$
$$= \sum_i P(B \mid A_i)P(A_i)$$

The probability of B is the sum of the probability of B under each case, weighted by that case's probability.

Bayes' theorem

$$P(A \mid B) = P(B \mid A)P(A) / P(B)$$

Posterior = likelihood × prior / evidence.

Naive Bayes

An efficient machine-learning approach used primarily for classification.

It applies Bayes' theorem and assumes the input features are conditionally independent given the class. Bayes' theorem itself does not require this assumption.

For several features x_1, ..., x_m:

$$P(C \mid x_1,...,x_m) \propto P(C) \times \prod_j P(x_j \mid C)$$

Compute the posterior score for each class (such as Yes or No), then select the class with the highest score.

Example question: given weather data, should a player play on a sunny day? The frequency table and calculation follow.

Bayes' theorem · weather example

Frequency table

Weather	Play: Yes	Play: No	Total
Overcast	5	0	5
Rainy	2	2	4
Sunny	3	2	5
Total	10	4	14

Find P(Yes | Sunny): the probability of playing given a sunny day.

Play: Yes

$$P(\text{Yes} \mid \text{Sunny})$$

$$= P(\text{Sunny} \mid \text{Yes})P(\text{Yes}) / P(\text{Sunny})$$

From the table:

$$P(\text{Sunny} \mid \text{Yes}) = 3/10 = 0.30$$

$$P(\text{Yes}) = 10/14 \approx 0.7143$$

$$P(\text{Sunny}) = (3+2)/14 = 5/14 \approx 0.3571$$

Therefore:

$$P(\text{Yes} \mid \text{Sunny}) = (3/10 \times 10/14) / (5/14)$$

$$= 3/5 = 0.60$$

Play: No

$$P(\text{No} \mid \text{Sunny})$$

$$= P(\text{Sunny} \mid \text{No})P(\text{No}) / P(\text{Sunny})$$

$$P(\text{Sunny} \mid \text{No}) = 2/4 = 0.50$$

$$P(\text{No}) = 4/14 \approx 0.2857$$

$$P(\text{No} \mid \text{Sunny}) = (2/4 \times 4/14) / (5/14)$$

$$= 2/5 = 0.40$$

Decision

Since 0.60 > 0.40, predict Yes: the player can play on a sunny day.

Using exact fractions avoids rounding errors. The two posterior probabilities sum to 1.

Random variables & expected value

A random variable assigns a numerical value to each outcome of a random experiment. Its domain is the sample space and its values are real numbers.

Discrete or continuous?

Discrete: a finite or countably infinite set of possible values. Examples: number of defective pieces, number of steps.

Continuous: values in an interval or collection of intervals. Measurements such as time, weight, volume and length are typical examples.

Probability mass function (discrete)

$$p(x) = P(X = x)$$

Each probability must be nonnegative, and all probabilities must sum to 1.

x	1	2	3	4
Valid p(x)	0.25	0.25	0.25	0.25
Invalid p(x)	0.40	−0.10	0.30	0.40

The second row of probabilities is invalid because it includes a negative value, even though its sum is 1.

Expected value

The mean of a random variable:

$$E(X) = \mu = \Sigma_x \, x \, p(x)$$

Variance & standard deviation

Variance summarizes variability in the possible values:

$$Var(X) = \sigma^2 = \Sigma_x \, (x - \mu)^2 \, p(x)$$

Shortcut formula:

$$Var(X) = E(X^2) - [E(X)]^2$$

$$= \Sigma_x \, x^2 p(x) - \mu^2$$

Standard deviation is the positive square root of variance.

Properties of expectation

For constants a and b:

$$E(aX) = a\,E(X)$$

$$E(X+b) = E(X)+b$$

$$E(aX+b) = a\,E(X)+b$$

Variance properties & a discrete example

Useful properties

For constants a and b:

$$\mathrm{Var}(aX) = a^2\,\mathrm{Var}(X)$$

$$\mathrm{Var}(X+b) = \mathrm{Var}(X);\ \mathrm{Var}(a) = 0$$

$$\mathrm{SD}(aX) = |a|\,\mathrm{SD}(X);\ \mathrm{SD}(X+b) = \mathrm{SD}(X)$$

Distribution checks

Discrete: $0 \le p(x) \le 1$ and $\Sigma p(x) = 1$.

Continuous: $f(x) \ge 0$ and $\int f(x)\,dx = 1$ over its support. A density may exceed 1; it is its area that represents probability.

Example

Given the following discrete distribution, compute $E(X)$, $\mathrm{Var}(X)$ and SD, then check variance with the shortcut.

x	1	2	4	8	16
$p(x)$	0.05	0.10	0.35	0.40	0.10

The probabilities sum to 1.

x	$p(x)$	$xp(x)$	$x^2p(x)$
1	0.05	0.05	0.05
2	0.10	0.20	0.40
4	0.35	1.40	5.60
8	0.40	3.20	25.60
16	0.10	1.60	25.60
Total	1.00	6.45	57.25

$$\mu = \Sigma xp(x) = 6.45$$

Squared deviations, weighted by their probabilities:

x	$(x-6.45)^2$	$(x-6.45)^2p(x)$
1	29.7025	1.485125
2	19.8025	1.980250
4	6.0025	2.100875
8	2.4025	0.961000
16	91.2025	9.120250

The total weighted squared deviation is 15.6475. The calculation continues on the next page.

Continuous densities · worked example

Discrete example · results

$$E(X) = 6.45$$

$$\mathrm{Var}(X) = 57.25 - 6.45^2 = 15.6475$$

$$\mathrm{SD}(X) = \sqrt{15.6475} \approx 3.9557$$

These agree with the weighted squared-deviation calculation.

A continuous density

Suppose $f(x) = cx^2$ for $-1 \le x \le 1$, and $f(x) = 0$ otherwise.

The interval is the support of the distribution. Use integrals, rather than sums, for continuous variables.

Integration reminders

$$\int x^n \, dx = x^{n+1}/(n+1) + C, \; n \neq -1$$

$$\int e^{ax} \, dx = e^{ax}/a + C, \; a \neq 0$$

Integration by parts:

$$\int u \, dv = uv - \int v \, du$$

$$\int [1/(1+x^2)] \, dx = \arctan(x) + C$$

When evaluating a definite integral, substitute the upper and lower limits and subtract.

Exercise

Let X have density:

$$f(x) = cx^2, \; -1 \le x \le 1$$

$$f(x) = 0, \text{ otherwise}$$

Find:

(a) The normalizing constant c.

(b) $E(X)$ and $\mathrm{Var}(X)$.

(c) $P(X \ge 1/2)$.

The full solution follows.

Continuous density · solution

(a) Find c

The total area under a density is 1:

$$\int_{-1}^{1} cx^2\,dx = 1$$

$$c\left[x^3/3\right]_{-1}^{1} = c(1/3 - (-1/3)) = 1$$

$$2c/3 = 1 \rightarrow c = 3/2$$

Thus $f(x) = (3/2)x^2$ on $[-1,1]$, and 0 elsewhere.

(b) Mean & variance

$$E(X) = \int_{-1}^{1} x\,f(x)\,dx$$

$$= (3/2)\int_{-1}^{1} x^3\,dx$$

$$= \left[(3/8)x^4\right]_{-1}^{1} = 0$$

An odd integrand over symmetric limits integrates to zero.

$$E(X^2) = \int_{-1}^{1} x^2 f(x)\,dx$$

$$= (3/2)\int_{-1}^{1} x^4\,dx$$

$$= \left[(3/10)x^5\right]_{-1}^{1} = 3/5$$

$$Var(X) = E(X^2) - [E(X)]^2 = 3/5$$

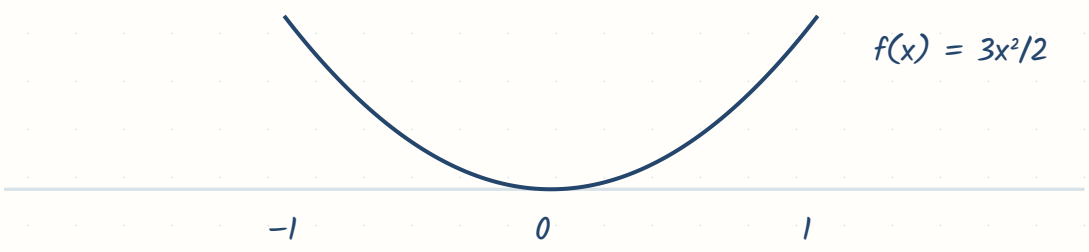
(c) Probability above 1/2

Change the integration limits to the event of interest:

$$P(X \geq 1/2) = \int_{1/2}^{1} (3/2)x^2\,dx$$

$$= \left[x^3/2\right]_{1/2}^{1}$$

$$= 1/2 - 1/16 = 7/16 = 0.4375$$



Densities, CDFs & joint distributions

Properties of a probability density

$$f(x) \geq 0;\ \int_{-\infty}^{+\infty} f(x)\,dx = 1$$

$$P(a \leq X \leq b) = \int_a^b f(x)\,dx$$

Probability is the area under the density from a to b.

Cumulative distribution function (CDF)

$$F(x) = P(X \leq x) = \int_{-\infty}^{x} f(t)\,dt$$

Where F is differentiable: $f(x) = dF(x)/dx$.

Continuous moments

$$\mu = E(X) = \int_{-\infty}^{+\infty} x\,f(x)\,dx$$

$$Var(X) = \int_{-\infty}^{+\infty} (x-\mu)^2 f(x)\,dx$$

$$= \int_{-\infty}^{+\infty} x^2 f(x)\,dx - \mu^2$$

$$SD(X) = \sigma = \sqrt{Var(X)}$$

Joint probability distribution · discrete

Use two random variables instead of one. Let X take values $x_1,\ldots,x_n$ and Y take values $y_1,\ldots,y_m$.

$$p_{ij} = P(X=x_i,\ Y=y_j)$$

The conditions are $0 \leq p_{ij} \leq 1$ and $\Sigma_i \Sigma_j\, p_{ij} = 1$.

Y / X	1	2	3	Row total
0	0.10	0.10	0.20	0.40
1	0.20	0.10	0.30	0.60
Column total	0.30	0.20	0.50	1.00

Marginal distributions come from summing over the other variable:

$P(X=1,2,3) = 0.30,\ 0.20,\ 0.50$.

$P(Y=0,1) = 0.40,\ 0.60$.

Joint densities, Bernoulli & binomial

Continuous joint distribution

A joint density must satisfy $f(x,y) \geq 0$ and:

$$\iint f(x,y)\,dx\,dy = 1$$

Integrate over a region R to find $P((X,Y) \in R)$. A density need not be bounded above by 1.

Bernoulli distribution

One trial with two outcomes: success (1) and failure (0).

Let $p = P(\text{success})$, $q = 1-p$.

$$P(X=x) = p^x q^{1-x},\ x = 0,1$$

It is zero for other values.

$$E(X) = 0\cdot q + 1\cdot p = p$$

$$E(X^2) = p;\ \mathrm{Var}(X) = p - p^2 = pq$$

Binomial distribution

Counts the number of successes in n independent Bernoulli trials.

Assumptions: n is fixed; trials are independent; each trial has two mutually exclusive outcomes; success probability p is constant.

$$P(X=x) = C(n,x)\,p^x\,q^{n-x},\ x = 0,\ldots,n$$

$$C(n,x) = n!\ /\ [x!(n-x)!]$$

$$\mu = np;\ \sigma^2 = npq$$

Die example

Probability of rolling a 4 on a fair die: $p = 1/6$. Probability of not rolling a 4: $q = 5/6$.

For 10 independent rolls, X counts the number of 4s.

$$P(X=10) = C(10,10)(1/6)^{10}(5/6)^0$$

$$= (1/6)^{10}$$

The binomial distribution is discrete.

Poisson & normal distributions

Poisson distribution

Models counts of events in a fixed interval under a constant-rate, independent-event model.

$$P(X=x) = e^{-\lambda}\lambda^{x} / x!,\ x = 0,1,2,\ldots$$

The probability is zero elsewhere; $\lambda > 0$.

$$E(X) = \lambda;\ Var(X) = \lambda$$

It is useful for rare events. A binomial distribution can be approximated by a Poisson distribution when n is large, p is small, and $\lambda = np$ stays moderate.

In the limiting argument: $n \to \infty$, $p \to 0$, while $np = \lambda$ stays constant.

Normal distribution

A continuous distribution used to model many approximately bell-shaped measurements, such as heights or measurement errors.

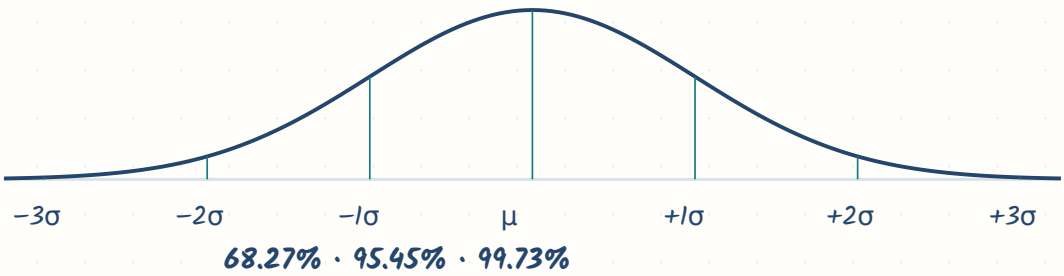
Write $X \sim N(\mu, \sigma^2)$, where $-\infty < \mu < \infty$ and $\sigma > 0$.

$$f(x) = [1/(\sigma\sqrt{(2\pi)})]\ \exp[-(x-\mu)^2/(2\sigma^2)]$$

The support is the entire real line: $-\infty < x < \infty$.

Standardization:

$$z = (X - \mu) / \sigma$$



68.27% · 95.45% · 99.73%

Empirical rule

About 68.27% of values lie within $\mu \pm \sigma$.

About 95.45% lie within $\mu \pm 2\sigma$.

About 99.73% lie within $\mu \pm 3\sigma$.

From the mean to 1σ: about 34.13% on each side; from 1σ to 2σ: 13.59%; from 2σ to 3σ: 2.14%.

Normal approximation & sampling

Properties of the normal curve

Bell-shaped and symmetric about its mean.

Mean = median = mode. Total area under the curve = 1.

The tails approach the horizontal axis but never touch it.

Normal approximation to a binomial

For sufficiently large np and nq (the notes use values above 5), approximate $X \sim \text{Binomial}(n,p)$ by $Y \sim N(np, npq)$.

$$Z = (Y - np) / \sqrt{(npq)}$$

Use a continuity correction when converting integer counts to continuous intervals.

Example · n = 40, p = q = 0.5

Then $\mu = 20$ and $\sigma = \sqrt{10} \approx 3.1623$.

$$P(X=12) \approx P(11.5 < Y < 12.5)$$

$$= P(-2.688 < Z < -2.372)$$

$$\approx 0.0051$$

Find the difference between the two normal CDF values, using a Z table or calculator.

For a cumulative event:

$$P(X \leq 12) \approx P(Y < 12.5)$$

Sampling

Selecting a group from a population. Example: select marks of 10 students from a class of 40.

Random sampling: select using a random procedure.

Stratified sampling: divide the population into groups (strata), then sample within each group.



Sampling variability occurs even with good random sampling. Biased selection can introduce additional error.

Sampling distributions & the CLT

Sampling variability

Different samples give different statistics. Larger random samples generally have lower standard errors.

For independent observations with population mean μ and SD σ:

$$E(\bar{x}) = \mu;\ SE(\bar{x}) = \sigma/\sqrt{n}$$

$$z = (\bar{x} - \mu) / (\sigma/\sqrt{n})$$

Number of samples

From a population of size N, for samples of size n:

With replacement, ordered draws: N^n possibilities.

Without replacement, unordered samples: $C(N,n)$.

Finite population correction

For simple random sampling without replacement from a finite population, use:

$$FPC = \sqrt{[(N-n)/(N-1)]}$$

$$SE(\bar{x}) = (\sigma/\sqrt{n}) \times FPC$$

The correction matters when the sampling fraction n/N is appreciable; a common rule is $n/N > 0.05$.

Central limit theorem

Under suitable independence and finite-variance conditions, the sampling distribution of the mean approaches normality as n grows.

The notes use $n \geq 30$ as a practical rule of thumb; the required size depends on the population distribution.

If the population itself is normal, the sample mean is normal for any n (for independent normal observations).

Use the standardized sample mean above; include FPC when applicable.

Sampling proportion

A sample proportion describes a group within a sample:

$$\hat{p} = x/n$$

Example: 10 red balls in a sample of 50 → $\hat{p} = 10/50 = 0.20$.

Proportions & confidence intervals

Sampling distribution of a proportion

For independent Bernoulli observations with $q = 1-p$:

$$E(\hat{p}) = p;\ SE(\hat{p}) = \sqrt{(pq/n)}$$

$$z = (\hat{p} - p) / \sqrt{(pq/n)}$$

Confidence interval · population mean

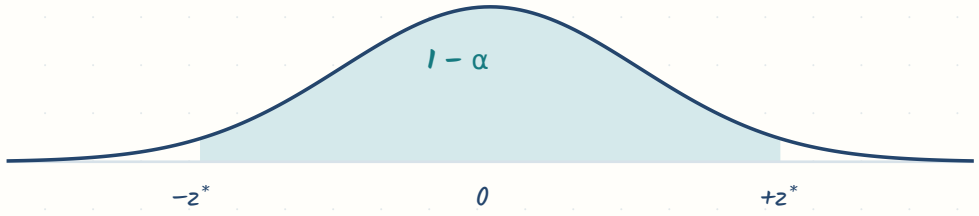
An interval estimate gives a range of plausible parameter values rather than a single point estimate.

For known population SD σ and an appropriate normal model:

$$\bar{x} - z^* \sigma/\sqrt{n} \le \mu \le \bar{x} + z^* \sigma/\sqrt{n}$$

Here $z^* = z_{1-\alpha/2}$, the positive normal critical value, and the confidence level is $1-\alpha$.

The combined area in the two tails is α; each tail has area $\alpha/2$.



For 95% confidence: $\alpha = 0.05$; $\alpha/2 = 0.025$; $z^* \approx 1.96$.

Confidence	z^* (two-sided)
90%	1.645
92%	1.751
95%	1.960
99%	2.576

For sampling without replacement from a finite population, multiply the standard error by $\sqrt{[(N-n)/(N-1)]}$ when appropriate.

Confidence interval · population proportion

An approximate large-sample interval is:

$$\hat{p} \pm z^* \sqrt{[\hat{p}(1-\hat{p})/n]}$$

The parameter being estimated is p, not μ.

Check the success/failure counts; a common minimum is $n\hat{p} \ge 5$ and $n(1-\hat{p}) \ge 5$. This is the basic normal (Wald) interval and can perform poorly near 0 or 1.

Hypothesis testing · foundations

A hypothesis is a statement or claim about a population. A statistical test assesses evidence from sample data.

Null & alternative hypotheses

H_0: the null hypothesis, often no change or no difference.

H_1 (or H_a): the alternative claim.

Examples: H_0: $\mu = 100$ versus H_1: $\mu \neq 100$.

A one-sided example: H_0: $\mu_1 \leq \mu_2$ versus H_1: $\mu_1 > \mu_2$.

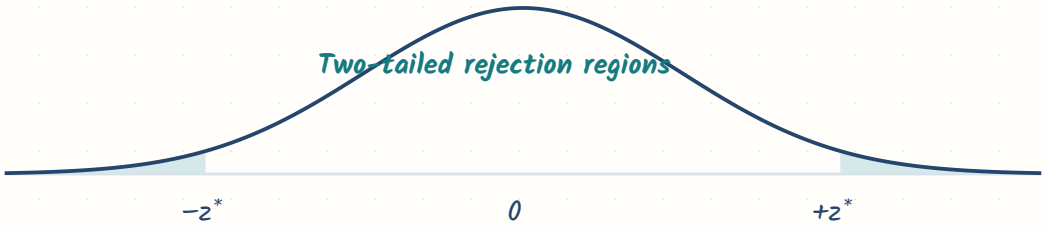
The null contains equality ($=$, $\leq$ or $\geq$). The alternative uses $\neq$, $<$ or $>$.

One-tailed & two-tailed tests

H_1: $\mu_1 < \mu_2$ → left-tailed test.

H_1: $\mu_1 > \mu_2$ → right-tailed test.

H_1: $\mu_1 \neq \mu_2$ → two-tailed test.



Critical region

The rejection region contains values of the test statistic that are sufficiently extreme under H_0.

The significance level α is the probability of a Type I error, assuming H_0 is true.

Errors in decisions

$$\alpha = P(\text{reject } H_0 \mid H_0 \text{ true})$$

$$\beta = P(\text{fail to reject } H_0 \mid H_0 \text{ false})$$

Decision	H_0 true	H_0 false
Fail to reject H_0	Correct	Type II error (β)
Reject H_0	Type I error (α)	Correct

Power = $1-\beta$. Failing to reject H_0 does not prove H_0 is true.

Hypothesis tests · formula sheet

Z tests

Used for one mean, a difference of means, one proportion, or a difference of proportions under the relevant model and large-sample conditions.

For means, known population standard deviations give Z statistics; large-sample estimated-SE versions are approximations.

One mean, null mean μ_0:

$$z = (\bar{x} - \mu_0) / (\sigma/\sqrt{n})$$

Two independent means, null difference Δ_0:

$$z = [(\bar{x}_1 - \bar{x}_2) - \Delta_0] / \sqrt{(\sigma_1^2/n_1 + \sigma_2^2/n_2)}$$

One proportion, null proportion p_0 and $q_0 = 1 - p_0$:

$$z = (\hat{p} - p_0) / \sqrt{(p_0 q_0 / n)}$$

Two proportions, H_0: $p_1 = p_2$, with pooled $\hat{p} = (x_1 + x_2)/(n_1 + n_2)$:

$$z = (\hat{p}_1 - \hat{p}_2) / \sqrt{[\hat{p}(1-\hat{p})(1/n_1 + 1/n_2)]}$$

For estimating a difference, the unpooled SE is $\sqrt{[\hat{p}_1(1-\hat{p}_1)/n_1 + \hat{p}_2(1-\hat{p}_2)/n_2]}$.

t tests · estimated standard deviations

One mean:

$$t = (\bar{x} - \mu_0) / (s/\sqrt{n}); \; df = n - 1$$

Two independent means, Welch's statistic:

$$t = [(\bar{x}_1 - \bar{x}_2) - \Delta_0] / \sqrt{(s_1^2/n_1 + s_2^2/n_2)}$$

Use Welch's degrees of freedom, or an appropriate pooled test if equal variances are assumed.

Paired observations, differences d_i:

$$t = (\bar{d} - \mu_{d,0}) / (s_d/\sqrt{n}); \; df = n - 1$$

Normal critical values

Significance α	1%	5%	10%
Two-tailed: $\pm z^*$	±2.576	±1.960	±1.645
One-tailed magnitude	2.326	1.645	1.282

p-values, t tests & chi-square tests

p-value

Under H_0, the p-value is the probability of a test statistic at least as extreme as the observed statistic, in the direction(s) specified by H_1.

Given $Z = 1.65$, the right-tail probability is about 0.0495. For a two-sided test, it is about 0.0989.

Reject H_0 when $p \leq \alpha$. Smaller p-values indicate greater incompatibility with the null model; they are not the probability that H_0 is true.

The original notes use these rough descriptive bands:

Below 1%: very strong evidence; 1–5%: strong evidence; 5–10%: weaker evidence; above 10%: little evidence against H_0.

Student's t test

Useful when σ is unknown and estimated by s, especially for small samples from an approximately normal population.

$$t = (\bar{x} - \mu_0) / (s/\sqrt{n})$$

Compare against a t table using the appropriate tail(s) and $n-1$ degrees of freedom for a one-sample test.

Chi-square tests

For categorical frequency data: goodness of fit or independence in a contingency table.

$$\chi^2 = \Sigma_i \, (O_i - E_i)^2 / E_i$$

O_i is the observed count; E_i is the expected count under H_0.

For independence: H_0 says the row and column variables are independent; H_1 says they are associated.

For r×c tables:

$$E_{ij} = (\text{row total} \times \text{column total}) / \text{grand total}$$

$$df = (r-1)(c-1)$$

For goodness of fit with k categories and no estimated parameters: $df = k-1$. Subtract the number of fitted parameters when required.

Use independent counts and check that expected frequencies are adequate for the approximation.

Chi-square · worked examples

1 · Accident counts by weekday

Test whether accidents are equally likely on each day (goodness of fit).

Day	Observed O	Expected E	$(O-E)^2/E$
Monday	184	150	7.7067
Tuesday	148	150	0.0267
Wednesday	145	150	0.1667
Thursday	153	150	0.0600
Friday	150	150	0.0000
Saturday	154	150	0.1067
Sunday	116	150	7.7067
Total	1050	1050	15.7733

$E = 1050/7 = 150$ for every day.

At $\alpha = 0.05$, $df = 7-1 = 6$; the critical χ^2 value is about 12.592.

Since $15.773 > 12.592$, reject equal weekday probabilities. The counts provide evidence that accident frequency varies by weekday.

2 · Smoking & cancer · independence

Group	Cancer	No cancer	Total
Smoker	400	300	700
Non-smoker	300	500	800
Total	700	800	1500

Expected counts under independence:

Group	Cancer	No cancer
Smoker	326.6667	373.3333
Non-smoker	373.3333	426.6667

Example: $E(\text{smoker, cancer}) = 700\times700/1500 = 326.6667$.

$$\chi^2 = \Sigma\,(O-E)^2/E \approx 57.8763;\ df = 1$$

For a 2×2 table $[a\ b;\ c\ d]$, a shortcut is:

$$\chi^2 = n(ad-bc)^2 / (r_1 r_2 c_1 c_2)$$

Here r_1, r_2 are row totals, c_1, c_2 are column totals, and n is the grand total.

Exact tests, variances & covariance

Fisher's exact probability · 2x2 table

For fixed row and column totals:

$$P(\text{table}) = (r_1!\ r_2!\ c_1!\ c_2!) / (n!\ a!\ b!\ c!\ d!)$$

A Fisher test p-value sums probabilities of tables as or more extreme according to the chosen alternative; it is not always this single-table probability.

Yates' continuity correction

For a 2×2 chi-square calculation, the corrected contribution is:

$$(|O-E| - 0.5)^2 / E$$

One population variance

For a normal population, test H_0: $\sigma^2=\sigma_0^2$ using:

$$\chi^2 = (n-1)s^2 / \sigma_0^2;\ df = n-1$$

The alternative may be $\sigma^2<\sigma_0^2$, $\sigma^2>\sigma_0^2$ or $\sigma^2\neq\sigma_0^2$; choose the matching tail(s).

Ratio of two variances · F test

For independent normal samples, test H_0: $\sigma_1^2=\sigma_2^2$:

$$F = s_1^2/s_2^2$$

Use an F distribution with numerator df n_1-1 and denominator df n_2-1. Read critical values from the F table for the chosen alternative and significance level.

Correlation analysis · covariance

Studies the existence and extent of a relationship between two variables.

Sample covariance:

$$\text{cov}(X,Y) = \sum_i (x_i-\bar{x})(y_i-\bar{y}) / (n-1)$$

Positive covariance: values tend to increase together.

Negative covariance: as one increases, the other tends to decrease.

Covariance describes co-variation and carries units. Zero covariance does not generally imply independence.

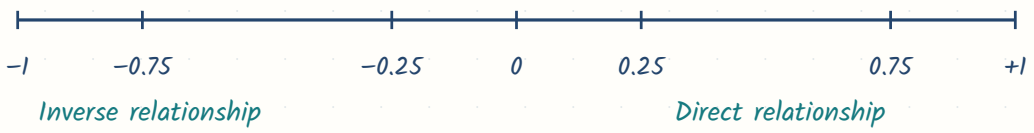
Correlation, regression & estimation

Pearson correlation coefficient

Remove the units of covariance by dividing by the standard deviations:

$$r = \Sigma(x_i-\bar{x})(y_i-\bar{y})$$
$$/\ \sqrt{[\Sigma(x_i-\bar{x})^2 \times \Sigma(y_i-\bar{y})^2]}$$

Equivalently $r = \mathrm{cov}(X,Y)/(s_X s_Y)$. The range is -1 to $+1$.



Negative r: inverse linear relationship. Positive r: direct linear relationship. Values near zero indicate weak linear association.

The original sketch marks $|r|$ near 0.25 as weak and near 0.75 as strong; interpretation depends on context.

Regression analysis

Simple linear regression fits a line to predict a response:

$$\hat{y} = \beta_0 + \beta_1 x$$

Given x and y observations, solve the normal equations for the fitted intercept and slope:

$$\Sigma y = n\beta_0 + \beta_1 \Sigma x$$
$$\Sigma xy = \beta_0 \Sigma x + \beta_1 \Sigma x^2$$

Maximum likelihood estimation · reminders

Bernoulli success probability: $\hat{p} = k/n$, where k is the number of successes.

Binomial(n,p), with known n and repeated observations: $\hat{p} = \bar{x}/n$.

Poisson rate: $\hat{\lambda} = \bar{x}$.

Normal mean and variance:

$$\hat{\mu} = \Sigma x_i/n = \bar{x}$$
$$\hat{\sigma}^2 = \Sigma(x_i-\bar{x})^2/n$$

The MLE variance uses n rather than the unbiased sample-variance denominator $n-1$.

For a continuous uniform distribution with both endpoints unknown, the MLE endpoints are the sample minimum and maximum.